Qihua Dong (He/Him/His)

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EDUCATION

Northeastern University, Boston, USA Sep 2023 – Present

PhD in Computer Engineering (Advisor: Prof. Raymond Fu); GPA: 3.96/4.00

Research interests: Multi-modal large language models, computer vision, artificial intelligence.

City University of Hong Kong, Hong Kong SAR, China Sep 2017 – Jul 2021

BSc in Computer Science with Mathematics Minor; GPA: 3.89/4.3 (Top 10%)

Selected coursework: Mathematical Analysis; Enhanced Calculus and Linear Algebra; Discrete Mathematics; Statistic Processes; Data Structures and Advanced Programming; Design and Analysis of Algorithms.

RESEARCH INTERESTS

Visual reasoning and multimodal LLMs; current focus on visual agents, tool use, and reinforcement learning for LLMs; prior work in weakly supervised learning, segmentation, and medical imaging.

PUBLICATIONS

Preprints and Under Review

- **Qihua Dong**, Gozde Sahin, Pei Wang, Zhaowei Cai, Robik Shrestha, Hao Yang, Davide Modolo. "Visual Reasoning through Tool-supervised Reinforcement Learning." *Submitted to CVPR*, 2025.
- **Qihua Dong**, Luis Figueroa, Handong Zhao, Kushal Kafle, Jason Kuen, Zhihong Ding, Scott Cohen, Yun Fu. "CoT Referring: Chain-of-Thought Grounding Improve Localization in Referring Expression Tasks." *Submitted to ICLR*, 2025. arXiv:2510.06243
- **Qihua Dong**, Yang Kuo, Ju Lin, Handong Zhao, Yitian Zhang, Yizhou Wang, Huimin Zeng, Jianglin Lu, Yun Fu. "Ref-Adv: Exploring MLLM Visual Reasoning in Referring Expression Tasks." *Submitted to ICLR*, 2025. OpenReview

Peer-Reviewed Publications

**: Equal contribution.*

- Hao Du*, **Qihua Dong***, Yan Xu, Jing Liao. "TDFormer: Top-Down Token Generation for 3D Medical Image Segmentation." *IEEE JBHI*, 2025. DOI:10.1109/JBHI
- **Qihua Dong**, Hao Du, Ying Song, Yan Xu, Jing Liao. "Preserving Tumor Volumes for Unsupervised Medical Image Registration." *ICCV*, 2023. arXiv:2309.10153
- Ruozhen He*, **Qihua Dong***, Jiayin Lin, Rynson Lau. "Weakly-Supervised Camouflaged Object Detection with Scribble Annotations." *AAAI*, 2023. arXiv:2207.14083
- Hao Du*, **Qihua Dong***, Yan Xu, Jing Liao. "Weakly-Supervised 3D Medical Image Segmentation using Geometric Prior and Contrastive Similarity." *IEEE TMI*, 2023. arXiv:2302.02125
- Yizhou Wang, Lingzhi Zhang, Yue Bai, Mang Tik Chiu, Zhengmian Hu, Mingyuan Zhang, **Qihua Dong**, Yu Yin, Sohrab Amirghodsi, Yun Fu. "Cautious Next Token Prediction." *ACL*, 2025. arXiv:2507.03038

Academic Service

- Outstanding Reviewer, ICCV 2025.

INDUSTRY EXPERIENCE

Amazon, AGI Foundation Team, WA, USA May 2025 – Present

Applied Scientist Intern (Mentors: Hao Yang, Zhaowei Cai, Pei Wang)

Built multi-tool visual agents (zoom, draw-box, rotate) with GRPO-based reasoning. Created curriculum

RL data pipelines for 7B/32B models with novel reward design, improving performance across chart, document, and high-resolution benchmarks.

Adobe Research, CA, USA May 2024 – Feb 2025

Research Intern (Mentors: Luis Figueroa, Handong Zhao, Scott Cohen)

Developed multimodal LLM for referring image segmentation with structured chain-of-thought reasoning. Created novel benchmarks with enriched annotations for complex referring expressions.

RESEARCH EXPERIENCE

Multi-modal LLM in Visual Understanding and Reasoning Oct 2023 – Present

Northeastern University (Advisor: Prof. Raymond Fu)

Building multimodal LLMs capable of visual grounding, spatial reasoning, and zero-shot generalization for intelligent agent development with minimal fine-tuning via in-context examples.

3D Medical Imaging May 2022 – Mar 2023

City University of Hong Kong (Advisors: Prof. Jing Liao, Prof. Yan Xu)

- Developed a volume-preserving registration network for tumor alignment with a task-specific loss.
- Designed a dynamic transformer for efficient medical image segmentation via foreground-aware tokenization.
- Proposed a weakly supervised segmentation framework combining geometric priors and contrastive pre-training, achieving 12.5% Dice improvement on benchmark datasets.

Weakly-Supervised Camouflaged Object Detection Dec 2021 – Jun 2022

City University of Hong Kong (Advisor: Prof. Rynson Lau)

Initiated the first weakly supervised formulation and dataset for camouflaged object detection with new consistency and feature losses, surpassing SOTA COD baselines by 11% MAE.

ADDITIONAL EXPERIENCE

Texas A&M University, TX, USA Oct 2022 – Jan 2023

Research Assistant (Remote) with Prof. Wenping Wang; contributed to computer vision and computer graphics projects focused on facial analysis.

City University of Hong Kong, Hong Kong SAR, China Aug 2021 – Aug 2023

Full-time Research Assistant with Prof. Jing Liao; advanced 3D medical image analysis and weakly supervised learning projects.

City University of Hong Kong, Hong Kong SAR, China Sep 2020 – Feb 2021

Part-time Student Research Assistant with Prof. Jianping Wang; researched scheduling schemes for self-driving systems.

HONORS AND SCHOLARSHIPS

- First Class Honors, City University of Hong Kong, 2021.
- Bronze Medal, ICPC Asia Nanchang Regional Contest, 2019.
- Dean's List, City University of Hong Kong, 2017–2021.
- Full Tuition Scholarship, City University of Hong Kong, 2017–2021.

SKILLS

Programming	Python, C++, Matlab, R, JavaScript, Java, CSS
Frameworks	PyTorch, NumPy, Detectron2, scikit-image, MMDetection, MONAI, scikit-learn
Languages	English: TOEFL 110 (Nov 2022); GRE 330 (Analytical Writing 3.5)